

Perioperative Anesthetic and Surgical Complications of the Nuss Procedure

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Objective: The Nuss procedure is a chest wall remodeling surgery performed in patients with pectus excavatum. This study was performed to analyze perioperative surgical and anesthetic complications with the Nuss procedures.

Design: A retrospective analysis.

Setting: An academic hospital.

Participants: Two hundred fourteen patients (children, adolescents, and adults) undergoing the Nuss procedure over 6 years.

Interventions: Patient age and sex, premorbid diseases, indications for surgery, patient position during the procedure, the length of surgery, time to hospital discharge, postoperative analgesia method, and the presence of perioperative complications were recorded.

Measurements: No mortality was observed. The overall complication rate was 18.7%, but the overall event rate was

42.6% (91 events in 40 patients). Intraoperative hypotension, tachycardia, and hypercapnia were the most common complications (4.7%), followed by postoperative ileus (3.2%), pneumothorax (right, left, or bilateral; 4.2%), lung parenchymal laceration (2.3%), and postoperative nausea and vomiting (2.3%). Two patients had an ulnar nerve palsy and 1 patient had a brachial nerve palsy as a result of surgical position.

Conclusion: Although the Nuss procedure is reported to be minimally invasive, some serious complications concerning both surgery and anesthesia should not be overlooked.
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KEY WORDS: pectus excavatum, chest wall deformity, Nuss procedure, Nuss operation, minimally invasive surgery, complication, anesthesia

THE NUSS PROCEDURE IS a chest wall remodeling surgery performed in adolescents suffering from pectus excavatum. The number of patients undergoing surgical correction with the Nuss procedure is increasing, and when compared with other traditional procedures such as chondrosternal resection, its primary advantage is that it is minimally invasive. Studies on the surgical management, postoperative complications, and surgical outcomes of the Nuss procedure are available, but the literature on its anesthetic management is limited to mostly case reports.¹⁻³ This retrospective study was performed to analyze the authors' institutional experience with the perioperative management and complications concerning both surgery and anesthesia in Nuss procedures.

MATERIALS AND METHODS

After obtaining ethical committee approval, a retrospective review of perioperative medical records of 214 patients (children, adolescents, and adults) who underwent the Nuss procedure for pectus excavatum repair at the authors' institute from August 2005 to June 2011 was performed. Age, sex, premorbid diseases, surgical indications, the position of the patient during the procedure, length of anesthesia and surgery, time to discharge from the hospital, postoperative analgesic methods, and the presence of perioperative complications were recorded. Patients were diagnosed as having postoperative nerve injury if they experienced paresthesia of the upper extremities.

There were no exclusion criteria. Medical history was recorded, physical examination was carried out, and complete blood count and posteroanterior and lateral chest radiographs were taken for preoperative evaluation. Echocardiography was performed for all patients, but computed tomography (CT) to determine the Haller index was not performed routinely. Forced expiratory volume in one second (FEV₁), forced vital capacity (FVC), and FEV₁/FVC were measured 1 day before and 6 months after surgery.

All patients were premedicated with 0.05 mg/kg of midazolam and 0.015 mg/kg of atropine sulfate intramuscularly. They were all administered a prophylactic antibiotic regimen (20 mg/kg of

cefazolin sodium IV) until hospital discharge. General anesthesia was induced with 3 to 5 mg/kg of thiopental sodium, and 0.6 mg/kg of rocuronium bromide. Anesthesia was maintained with 2% sevoflurane and 70% nitrous oxide in oxygen. All patients were intubated with a single-lumen tube and mechanically ventilated to maintain end-tidal carbon dioxide concentrations (ETCO₂) between 30 and 35 mmHg. Heart rate, peripheral oxygen saturation, and ETCO₂ were monitored continuously; non-invasive blood pressure readings were monitored at 5-minute intervals. Mean arterial blood pressure below 75% of the patient's normal value on 2 consecutive measurements was considered as hypotension, while an increase in heart rate more than 25% of the patient's normal value was considered as tachycardia. An increase in the ETCO₂ to 40 mmHg or more was diagnosed as hypercapnia.

The Nuss procedure was performed as follows: To facilitate surgical manipulation and to minimize cardiac or lung injury, a capnothorax with 5 to 15 mmHg of pressure was created by carbon dioxide insufflation into the right pleural cavity, compressing the ipsilateral lung. A horizontal line was drawn through the region with the deepest point of the deformity on the anterior chest wall. Two transverse incisions were made on both sides, at the anterior axillary line. Subcutaneous tunnels were created bilaterally by blunt dissection reaching the highest points of the deformity. Meanwhile, the chosen template was shaped on the chest, and the bar itself was bent to match the template. A separate 1-cm incision was made at the right midaxillary line to place the thoracoscope. A 5- or 10-mm blunt-tip

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Table 1. Demographic Data of the Patients, Duration of Surgery, Anesthesia, and Length of Hospitalization (Mean $\pm$ SD; Ranges), With Median Age Given

Age, y	17.9 $\pm$ 5.5 (6-35) (17)
Male/Female	184/30
Weight, kg	61.5 $\pm$ 14.2 (20-85)
Duration of surgery, min	72.9 $\pm$ 27.7 (30-180)
Duration of anesthesia, min	84 $\pm$ 28.3 (35-189)
Length of hospitalization, d	6.4 $\pm$ 2.5 (2-18)

trocars were introduced into the right pleural space and the videothoracoscope was inserted to explore the pleural cavity for the deepest point of the deformity and for any pleural adhesions. Following the exploration, the introducer was inserted through the right incision and forwarded to the deepest point of the deformity to create a tunnel between the sternum and the pericardium. When the retrosternal tunnel was created, the introducer was pushed through the left incision. Tape was tied to the eyelet of the introducer and pulled back, guiding the tape through the right thoracic cavity, and then out of the right incision. The bent bar was tied to the tape and pulled through the tunnel from right to left with the convex side facing down. The bar was rotated 180°, pushing the sternum upward. In most cases, 1 stabilizer was placed on the left side as close as possible to the exit point from the thoracic cavity to avoid rotation. Two stabilizers, 1 on each side, were placed when needed. The bar and the stabilizer were secured on the muscles by sutures. The air in the right pleural space was drained through a suction catheter, and the incisions were closed in layers with absorbable sutures. The sternum then was elevated. Ribs and costal cartilages were reformed. No drains were placed unless pneumonolysis for pleural adhesions had been performed.

The postoperative analgesia of the patients was maintained with 1 of the following methods for 48 hours after the surgery:

1. Intravenous patient-controlled analgesia (IV PCA) using a solution containing 10 mg/mL of meperidine, with a 1 or 1.5 mL bolus and 8 minutes lockout time, with a rescue analgesia of 0.5 mg/kg of meperidine or;
2. Thoracic epidural PCA using a solution containing 3 μ g/mL of fentanyl, 0.125% bupivacaine hydrochloride with 3 mL bolus, 15 minutes lockout time, and 2 mL/h of basal infusion, with a rescue analgesia of 15 mg of bupivacaine in 6 mL of normal saline, epidurally or;
3. Intermittent intravenous bolus doses of 1 mg/kg of meperidine.

In each of the analgesic methods, 20 mg/kg of paracetamol orally (4 times a day) also was administered.

In patients over 18 years of age, thoracic epidural catheters were inserted in the intervertebral space, Th 7-8, 30 minutes before surgery, under sedation. Needle placement (Touhy 18 G) in the epidural space was verified under fluoroscopy. Fifty micrograms of fentanyl and 0.25% bupivacaine in 8 mL were injected.

The numeric rating scale (NRS, with 0, no pain; 1-3, mild pain; 4-6, moderate pain; and 7-10, severe pain) was used to assess postoperative pain.

Patients were discharged from the hospital when they were devoid of any side effects.

For statistical analysis, GraphPad InStat version 3.00 for Windows 95 was used. Data were tabulated and simple descriptive statistics were performed. Results were reported as percentage or means, standard deviation, and ranges. Preoperative and postoperative spirometric data of the patients showing restrictive pulmonary diseases and postoperative NRS scores were compared with repeated measures of ANOVA,

with Tukey as a post-hoc test. Fisher's exact test was performed for comparison of the side effects of the analgesic methods; $p < 0.05$ was considered significant.

RESULTS

Demographic data of the patients, duration of surgery and anesthesia, and the length of hospitalization are shown in Table 1. The indication for surgery in all cases was bad cosmetic appearance as expressed by the patient. Preoperative echocardiographic evaluation revealed that 18 patients had a mild-to-moderate cardiac pathology; 10 patients had more than 1 pathology (Table 2).

Respiratory function tests were performed in only 188 patients; the rest of the patients were uncooperative and could not perform spirometry tests. Twenty-eight percent (53 patients) had FEV₁ and FVC less than 70% with a normal FEV₁/FVC, indicating a restrictive pulmonary dysfunction pattern. Respiratory function tests 6 months after the surgery revealed that only 11% of the patients were suffering from a restrictive pulmonary dysfunction (Table 3).

No mortality was observed. The overall complication rate was 18.7%, but the overall event rate was 42.6% (91 events in 40 patients with complications). Intraoperative hypotension, tachycardia, and hypercapnia were the most common complications (4.7%), followed by postoperative ileus (3.2%), pneumothorax (right, left, or bilateral) (4.2%), lung parenchymal laceration (2.3%), and postoperative nausea and vomiting (2.3%; Table 4).

Acute increases in ETCO₂ (46.4 $\pm$ 1.3 mmHg) and tachycardia (111.4 $\pm$ 4.9 beats/min) and decreases in mean arterial pressure (50.5 $\pm$ 1.7 mmHg) were observed together during carbon dioxide insufflation or bar insertion in 10 patients (4.7%).

The positions of the patients were as follows:

1. Supine position (3 patients).
2. Supine position with the arms suspended and secured to the ether screen (66 patients).
3. Supine position with the left arm in neutral position and right arm suspended and secured to the ether screen (145 patients; Fig 1).

The first 3 patients were placed in a neutral supine position. As 1 patient had a brachial plexus injury, the remaining 66 patients were positioned in a supine position with both arms

Table 2. Preoperative Echocardiographic Findings

Cardiac pathologies	No.
Mild mitral regurgitation	8
Mild tricuspid regurgitation	8
Mild aortic regurgitation	4
Mild pulmonary regurgitation	3
Mild tricuspid regurgitation	2
Moderate tricuspid regurgitation	2
Bicuspid aortic valve disease	2
Mild pulmonary regurgitation	1
Patent foramen ovale	1
Patent ductus arteriosus	1
Hypertrophic cardiomyopathy	1
Total	30

Table 3. Preoperative and Postoperative FEV₁ and FVC Values of the Patients With Restrictive Respiratory Disease

	Pre, %	Post, %	p Value
FEV ₁	64.6 ± 5.5 (48-69)	78.7 ± 8 (56-94)	p < 0.001
FVC	64.7 ± 3.8 (55-69)	78.1 ± 6.6 (56-79)	p < 0.001

Abbreviations: FEV₁, Forced expiratory volume in one second, FVC, forced vital capacity.

suspended to the ether screen. Two patients in this position had ulnar nerve palsy, and, therefore, the remaining 145 patients were placed in a supine position with the left arm in a neutral position, and to ease the manipulation of the thoracoscope, the right arm was suspended and secured to the ether screen (Fig 1). No nerve palsy developed in this position.

Only 1 patient required a transfusion of 600 mL of erythrocyte suspension. This patient had an iatrogenic myocardial laceration.

Postoperative analgesic methods used in patients were thoracic epidural PCA (31.8%), IV PCA (35%), and intermittent IV opioid administration (33.2%). The frequencies of side effects from different analgesic methods are presented in Table 5. Regarding side effects, no significant statistical difference was determined. Table 6 represents the NRS scores of the patients.

Table 4. Short- and Long-Term Complications of the Nuss Procedure Concerning Both Anesthesia and Surgery

Complications	No.	(%)
Pneumothorax	9	(4.2)
Parenchymal laceration	5	(2.3)
Nerve palsy	3	(1.4)
Pleural adhesion	2	(0.9)
Pleural effusion	3	(1.4)
Intercostal hemorrhage	2	(0.9)
Pericardial laceration	2	(0.9)
Pericarditis	1	(0.5)
Atelectasis	2	(0.9)
Cellulitis	4	(1.9)
Stabilizer displacement	1	(0.5)
Arrhythmia (SVT)	1	(0.5)
Myocardial laceration	1	(0.5)
Apneic episodes	1	(0.5)
Prolonged air leak	2	(0.9)
Diaphragmatic laceration	1	(0.5)
Thymic hemorrhage	1	(0.5)
Rib fracture	1	(0.5)
Intercostal muscle laceration	1	(0.5)
Parasternal hemorrhage	1	(0.5)
Horner syndrome	1	(0.5)
Intraoperative hypotension	10	(4.7)
Intraoperative tachycardia	10	(4.7)
Intraoperative hypercapnia	10	(4.7)
Postoperative nausea vomiting	5	(2.3)
Postoperative ileus	7	(3.2)
Postoperative pruritus	2	(0.9)
Postoperative urinary retention	2	(0.9)
Total	91	(42.6)

Abbreviation: SVT, supraventricular tachycardia.

**Fig 1. Ideal patient position for Nuss procedure.**

DISCUSSION

The authors' 6-year experience revealed that the Nuss procedure had a few serious complications concerning both surgery and anesthesia, such as myocardial and lung parenchymal lacerations, pneumothorax, and nerve injuries in the upper extremities.

The authors performed Pubmed Central and Google Scholar searches for the following key words: Pectus excavatum, chest wall deformity, Nuss procedure, Nuss operation, minimally invasive surgery, complication, and anesthesia. Regarding surgical management, many studies in the literature suggest that the Nuss procedure is a safe, short, and minimally invasive method leading to satisfactory cosmetic results.⁴⁻⁶ Few studies on the anesthetic management of the Nuss procedure exist in the literature and they are presented solely as case reports.^{1,2,7} Despite being limited by a retrospective design, this study described complications related to both anesthesia and surgery, in a large series of patients undergoing the Nuss procedure.

Haller et al⁸ introduced a preoperative CT index derived by dividing the transverse chest diameter by the anteroposterior diameter; they suggested that an index more than 3.25 correlated with a severe deformity. This index also has a good predictable value on the degree of reduced pulmonary function.⁹ The authors did not routinely perform CT scans in the patients, but spirometric measurements were made in cooperative patients to evaluate pulmonary dysfunction. Restrictive patterns

Table 5. Side Effects in Patients in Different Analgesic Groups

	Nausea/Vomiting	Ileus	Pruritus	Urinary Retention	Apneic Episodes
TE PCA (n = 68)	2 (3%)	1 (1.5%)	2 (3%)	2 (3%)	1 (1.5%)
IV PCA (n = 75)	3 (4%)	6 (8%)	2 (2.7%)	0	0
IV intermittent (n = 71)	5 (6.7%)	5 (6.7%)	2 (2.8%)	0	0

Abbreviation: TE PCA, thoracic epidural PCA.

in spirometry usually are seen in patients with pectus excavatum deformity.⁹ The authors also found that 28% of the patients had restrictive pulmonary dysfunction. Significant improvements regarding pulmonary functions occurred after the Nuss procedure.¹⁰ At the sixth month of the postoperative period, pulmonary function tests in the patients also were performed to assess whether or not they had improved, and the authors found that the percentage of patients with restrictive pulmonary dysfunction decreased to 11%.

In this study, the overall complication rate was 18.7%. Protopapas et al¹¹ reported an incidence of morbidity of 15.4%, in contrast to the findings of this study. In fact, as the authors took surgery and anesthesia-related complications into consideration, the complication rates were different from those reported in the literature.^{12,13} The most serious complication during the procedure was a myocardial laceration in 1 patient, resulting in significant blood loss. Pneumothorax incidence requiring chest tube insertion in the patients in this study (4.2%) was lower than that published in the literature.^{3,11} At the end of the procedure, the authors usually drained the residual air in the pleura via an aspiration catheter through the thoracoscopy incision, under positive-pressure ventilation.¹⁴ Variations between pneumothorax incidences in studies were probably due to the differences in diagnostic criteria. The authors consider that pneumothoraces observed in chest radiographs in the postoperative period usually resorb spontaneously without any intervention.

In Nuss procedures, the primary aim is to have good visualization and cause no harm to intrathoracic structures. Some centers, to obtain a better inspection field, prefer to use double-lumen endotracheal tubes to deflate the corresponding lung.³ An alternative is to use a single-lumen endotracheal tube with insufflation of carbon dioxide to compress the ipsilateral lung.¹⁴ Nevertheless, adverse pulmonary or hemodynamic effects may develop in Nuss procedures in which capnotherax is created.

The Nuss procedure is not a classic thoracoscopy. The main difference between thoracoscopic surgeries and the Nuss

procedure is that the Nuss procedure is performed in the supine position, eliminating the disadvantages of the lateral decubitus position of the classic thoracoscopic surgeries. Furthermore, hemodynamic effects of CO₂ insufflation during thoracoscopy may be more obvious when the patient is in the lateral decubitus position. Although one-lung ventilation usually is mandatory in thoracoscopies, it is not obligatory in Nuss procedures.

One-lung ventilation increases mechanical stress in the lung, affecting both ventilation and perfusion. It should be emphasized that both techniques have their own disadvantages. From a clinical point of view, the choice of either technique has to be made according to the preference and experience of the surgeon. In the authors' institution, thoracic surgeons prefer creating a capnotherax and ventilating the patient with a single-lumen endotracheal tube to obtain an easy manipulation field when performing Nuss procedures. Sudden hemodynamic and pulmonary instabilities such as tachycardia, hypotension, or acute increases in end-tidal carbon dioxide concentrations were observed in 4.7% of the patients either during carbon dioxide insufflation into the right pleural cavity or bar insertion. One case in the literature reported a sudden decrease in arterial blood pressure during the manipulation of the pectus bar in the anterior mediastinum due to compression of the heart.³ Acute hemodynamic and pulmonary deteriorations during capnotherax or bar insertion and sudden blood loss, as the authors experienced as a result of a myocardial laceration in 1 patient, are all hazardous complications during Nuss procedures. Although the Nuss procedure is reported to be minimally invasive, the authors suggest that the probability of acute hemodynamic deterioration necessitates the insertion of an intra-arterial catheter for continuous monitoring of the arterial pressure.

In the Nuss procedure in which thoracoscopy is used, it is necessary to place the patient in a supine position and in such a way that thoracoscopic maneuvers may be performed easily.^{14,15} The first 3 patients in this study were positioned in a supine position, and 1 of these patients had a brachial plexus injury. The reason for this may be explained as follows:

Table 6. NRS Scores of the Patients

	0 h	1 h	8 h	24 h	48 h
TE PCA	0.9 ± 0.8	0.9 ± 0.6	0.7 ± 0.7	0.6 ± 0.5	0.7 ± 0.7
IV PCA	6.1 ± 1.3*	1.7 ± 1*	1 ± 0.9	0.6 ± 0.6	0.9 ± 0.6
IV int	7.2 ± 1.4‡	3.4 ± 1§	3.5 ± 1.3§	1.7 ± 1.3	1.7 ± 1.2

Abbreviations: TE PCA, thoracic epidural PCA; IV Int, Intravenous intermittent.

* p ≤ 0.001 significantly different from the scores of other times.

† p ≤ 0.001 significantly different from the scores of 8.24 and 48 hours.

‡ p ≤ 0.001 significantly different from the scores of other times.

§ p ≤ 0.001 significantly different from the scores of 24 and 48 hours.

In this position, it was impossible to keep the upper extremities in a neutral position; thoracoscopic manipulations by the surgeons led to hyperextension of the arms, resulting in brachial plexus injury. To avoid this complication, the remaining 66 patients were again positioned in the neutral position, but with both arms suspended and secured to the ether screen. However, this position was not devoid of complications: 2 patients suffered from ulnar nerve palsy due to the compression of the suspension device. The remaining 145 patients were switched to a supine position with the left arm in a neutral position and the right arm suspended and secured to the ether screen. Patients suffered no nerve palsy in this position. Fox et al⁷ reported that 5.2% of their patients had a transient brachial plexus injury. The positions of their patients were different from those of the patients in the present study. They reported no ulnar nerve palsy because there was no pressure on the ulnar nerve.

Despite being a minimally invasive procedure, postoperative pain is a major problem after the Nuss procedure. An effective postoperative analgesia is essential to provide patient comfort and to avoid bar displacement and long bed rest. The authors used different postoperative analgesic methods in the patients in this study. An example of this diversity is that some of the patients were too young to tolerate the insertion of a thoracic

epidural catheter before the induction of anesthesia, which is the routine practice of the authors' institution. Although there was a lack of standardization, the authors still performed statistical analysis. Following the Nuss procedure, pain control with intermittent IV drug administration seemed to be unsatisfactory and, with IV PCA, ineffective in the early minutes of the postoperative period but satisfactory thereafter. In agreement with the findings of this study, thoracic epidural analgesia is suggested to be an efficient and safe method for reducing pain, providing early mobilization and improving pulmonary hygiene in the postoperative period.^{16,17} Nevertheless, the authors suggest that controlled and randomized studies are still lacking on this subject. With regard to pain in the Nuss procedure, it should be noted that postoperative pain in pediatric patients tends to occur in the anterior part of the thorax, contrary to adult patients.¹⁸ In addition, pain after this procedure mainly is related to the number of bars inserted during surgery.¹⁹

In conclusion, the Nuss procedure is considered a minimally invasive procedure from the surgical perspective. However, some serious complications concerning both surgery and anesthesia are probable. The risks should not be underestimated, and the authors strongly recommend vigilant follow-up during and after this procedure to avoid these troublesome complications.

REFERENCES

- Conti ME: Anesthetic management of acute subcutaneous emphysema and pneumothorax following a Nuss procedure: A case report. *AANA J* 77:208-211, 2009
- Ichizawa M, Morimura E, Tsuchiya N, et al: A case report of anesthetic management of the minimally invasive Nuss operation for pectus excavatum. *J Anesth* 19:73-74, 2005
- Futagawa K, Suwa I, Okuda T, et al: Anesthetic management for the minimally invasive Nuss procedure in 21 patients with pectus excavatum. *J Anesth* 20:48-50, 2006
- Kelly RE, Goretsky MJ, Obermeyer R, et al: Twenty-one years of experience with minimally invasive repair of pectus excavatum by the Nuss procedure in 1,215 patients. *Ann Surg* 252:1072-1081, 2010
- Kelly RE, Shamberger RC, Mellins RB, et al: Prospective multicenter study of surgical correction of pectus excavatum: Design, perioperative complications, pain, and baseline pulmonary function facilitated by internet-based data collection. *J Am Coll Surg* 205:205-216, 2007
- Engum S, Rescorla F, West K, et al: Is the grass greener? Early results of the Nuss procedure. *J Pediatr Surg* 35:246-251, 2000
- Fox ME, Bensard DD, Roaten JB, et al: Positioning for the Nuss procedure: Avoiding brachial plexus injury. *Paediatr Anaesth* 15: 1067-1071, 2005
- Haller JA Jr, Kramer SS, Lietman SA: Use of CT scans in selection of patients for pectus excavatum surgery: A preliminary report. *J Pediatr Surg* 22:904-906, 1987
- Lawson ML, Mellins RB, Paulson JF, et al: Increasing severity of pectus excavatum is associated with reduced pulmonary function. *J Pediatr* 159:256-261, 2011
- Lawson ML, Mellins RB, Tabangin M, et al: Impact of pectus excavatum on pulmonary function before and after repair with the Nuss procedure. *J Pediatr Surg* 40:174-180, 2005
- Protopapas AD, Athanasiou T: Perioperative data on the Nuss procedure in children with pectus excavatum: Independent survey of the first 20 years' data. *J Cardiothorac Surg* 3:40, 2008
- Croitou DP, Kelly RE Jr, Goretsky MJ, et al: Experience and modification update for the minimally invasive Nuss technique for pectus excavatum repair in 303 patients. *J Pediatr Surg* 37:437-445, 2002
- Castellani C, Schalamon J, Saxena AK, et al: Early complications of the Nuss procedure for pectus excavatum: A prospective study. *Pediatr Surg Int* 24:659-666, 2008
- Yüksel M, Bostancı K, Evman S: Minimally invasive repair after inefficient open surgery for pectus excavatum. *Eur J Cardiothorac Surg* 40:625-629, 2011
- Nuss D: Minimally invasive surgical repair of pectus excavatum. *Semin Pediatr Surg* 17:209-217, 2008
- Reinos-Barbero F, Fernández A, Durán P, et al: Thoracic epidural analgesia vs patient-controlled analgesia with intravenous fentanyl in children treated for pectus excavatum with the Nuss procedure. *Rev Esp Anestesiol Reanim* 57:214-219, 2010
- Walaszczyk M, Knapik P, Misiolek H, et al: Epidural and opioid analgesia following the Nuss procedure. *Med Sci Monit* 17:81-86, 2011
- Nagasao T, Miyamoto J, Ichihara K, et al: Age-related change of postoperative pain location after Nuss procedure for pectus excavatum. *Eur J Cardiothorac Surg* 38:201-208, 2010
- Nagaso T, Miyamoto J, Kokaji K, et al: Double-bar application decreases postoperative pain after the Nuss procedure. *J Thorac Cardiovasc Surg* 140:39-44, 2010